FET small-signal II

Answers and worked solutions · Day 20 · Thu 12 Nov · pack E12

5. Practice

From your course files

WS9 Q5 (course question).

Circuit (from Fig. 5): $V_{DD} = 22$ V. Gate on the divider 91 MΩ (top) / 11 MΩ (bottom) with a bypass capacitor to ground, so the gate is at ac ground. V_i is applied to the source, which has $R_S = 1$ kΩ to ground. $R_D = 2.2$ kΩ, output at the drain. $I_{DSS} = 7.5$ mA, $V_P = -4$ V, $r_d = 30$ kΩ. This is a common-gate amplifier.

DC.

$$V_G = 22 \cdot \frac{11}{91+11} = 2.373 \text{ V}, \qquad V_{GS} = V_G - I_D R_S = 2.373 - 1000 I_D$$

Substituting into $I_D = 7.5\text{ mA}\,(1 - V_{GS}/(-4))^2$ and solving the quadratic gives (the valid root, $V_P < V_{GS} < 0$):

$$I_{DQ} = 3.60 \text{ mA}, \qquad V_{GSQ} = -1.23 \text{ V}$$

Check: $V_{DS} = 22 - 3.60\text{ mA} \times 3.2\text{ k}\Omega = 10.48$ V, which is above $V_{GS} - V_P = 2.77$ V. Saturation holds.

g_m.

$$g_{m0} = \frac{2(7.5)}{4} = 3.75 \text{ mS}, \qquad g_m = 3.75\left(1 - \frac{-1.23}{-4}\right) = 2.60 \text{ mS}$$

> r_d **CHECK:** $r_d = 30$ kΩ and $10R_D = 22$ kΩ, so $r_d \geq 10R_D$ just holds. Use the full formulas anyway since r_d is given.

$$Z_i = R_S \parallel \frac{r_d + R_D}{1 + g_m r_d} = 1\text{ k}\Omega \parallel \frac{32.2\text{ k}\Omega}{1 + (2.60\text{ mS})(30\text{ k}\Omega)} = 1\text{ k}\Omega \parallel 408\ \Omega = \mathbf{290\ \Omega}$$

$$Z_o = R_D \parallel r_d = 2.2\text{k} \parallel 30\text{k} = \mathbf{2.05\ k\Omega}$$

$$A_v = \frac{g_m R_D + R_D/r_d}{1 + R_D/r_d} = \frac{(2.60\text{ mS})(2.2\text{ k}\Omega) + 0.0733}{1.0733} = \mathbf{5.39}$$

Positive, so V_o is in phase with V_i. Approximations: $Z_i \approx R_S \parallel 1/g_m = 278\ \Omega$ and $A_v \approx g_m R_D = 5.72$, within about 6%.

WS9 Q6 (course question).

"Match" means the CG input resistance equals the source resistance: $R_{in} = 1/g_m = R_{sig} = 100\ \Omega$, so

$$g_m = \frac{1}{100\ \Omega} = 10\ \text{mA/V}$$

With $V_{ov} = 0.20$ V and $g_m = 2I_D/V_{ov}$:

$$I_D = \frac{g_m V_{ov}}{2} = \frac{(10\ \text{mA/V})(0.20\ \text{V})}{2} = \mathbf{1\ mA}$$

Overall gain with $R_D = 2$ kΩ (total drain resistance):

$$G_v = \frac{R_{in}}{R_{in}+R_{sig}}\, g_m R_D = \frac{100}{200} \times (10\ \text{mS})(2\ \text{k}\Omega) = 0.5 \times 20 = \mathbf{10\ V/V}$$

Matching costs half the signal at the input. In course notation: $Z_i \approx 1/g_m$, $A_v \approx g_m R_D$, $A_{v_s} = \frac{Z_i}{Z_i+R_{sig}} A_v$.

WS9 Q7 (course question).

Fig. 6: $v_{sig} = 1$ V, $R_{sig} = 1$ MΩ, R_{in} very large, $R_o = 100\ \Omega$, $R_L = 1$ kΩ, $v_o \approx 0.9$ V.

R_{in} is very large (gate draws no current), so $v_i \approx v_{sig} = 1$ V.

The follower's output resistance is $R_o = 1/g_m$ (no R_S shown, r_o ignored), so $g_m = 1/100\ \Omega = 10$ mA/V. With $V_{ov} = 0.25$ V:

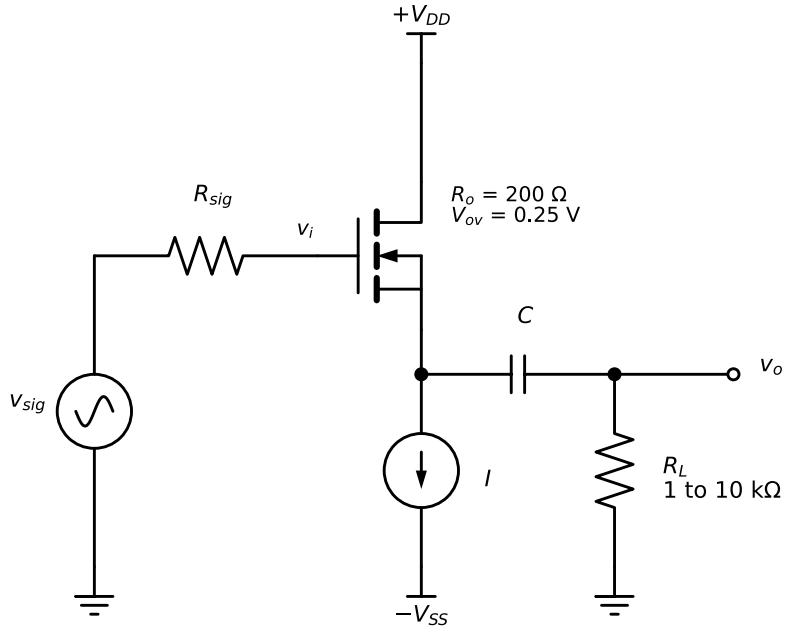
$$I_D = \frac{g_m V_{ov}}{2} = \frac{(10\ \text{mA/V})(0.25)}{2} = \mathbf{1.25\ mA}$$

Output: $v_o = v_i \dfrac{R_L}{R_L + 1/g_m} = 1 \times \dfrac{1000}{1100} = \mathbf{0.909\ V}$ (the figure's "≈ 0.9 V").

Gate–source signal: $v_{gs} = v_i - v_o = v_i \dfrac{1/g_m}{1/g_m + R_L} = \dfrac{100}{1100} = \mathbf{90.9\ mV}$.

Check: v_{gs} = 91 mV is well below $2V_{ov} = 0.5$ V, so the small-signal model is reasonable.

WS9 Q8 (course question).



WS9 Q8: source follower biased by a current source I

Here $k = 0.4$ mA/V² is the process parameter $\mu_n C_{ox}$, because W/L is the unknown. (If it were the course k, $g_m = 2kV_{ov} = 0.2$ mS and R_o would be 5 kΩ, not 200 Ω.)

$$R_o = \frac{1}{g_m} = 200\ \Omega \ \Rightarrow\ g_m = 5\ \text{mA/V}$$

$$\frac{W}{L} = \frac{g_m}{\mu_n C_{ox} V_{ov}} = \frac{5\ \text{mA/V}}{(0.4\ \text{mA/V}^2)(0.25\ \text{V})} = \mathbf{50}$$

$$I = I_D = \frac{g_m V_{ov}}{2} = \frac{(5)(0.25)}{2} = \mathbf{0.625\ mA}$$

Check: $I_D = \frac{1}{2}(0.4)(50)(0.25)^2 = 0.625$ mA. In course notation $k = \frac{1}{2}(0.4)(50) = 10$ mA/V² and $g_m = 2kV_{ov} = 5$ mS. Same answer.

Gain range ($G_v = R_L/(R_L + R_o)$, R_{sig} negligible against the gate):

$$R_L = 1\ \text{k}\Omega:\ G_v = \frac{1000}{1200} = 0.833, \qquad R_L = 10\ \text{k}\Omega:\ G_v = \frac{10\,000}{10\,200} = 0.980$$

So $\mathbf{0.83 \le G_v \le 0.98}$ V/V.

WS9 Q9 (course question).

Fig. 7: two identical self-bias stages. $V_{DD} = 18$ V, $R_D = 2.2$ kΩ, $R_S = 390\ \Omega$ bypassed (50 µF), $R_G = 10$ MΩ each, $V_i = 20$ mV. $I_{DSS} = 8$ mA, $V_P = -4.5$ V. r_d not given, so $r_d = \infty$.

DC (both stages). $V_{GS} = -390 I_D$. Solving with $I_D = 8\ \text{mA}(1 - V_{GS}/(-4.5))^2$ (valid root):

$$I_{DQ} = 3.70\ \text{mA}, \qquad V_{GSQ} = -1.44\ \text{V}$$

Check: $V_{DS} = 18 - 3.70\ \text{mA} \times 2.59\ \text{k}\Omega = 8.43$ V, above $V_{GS} - V_P = 3.06$ V.

g_m.

$$g_{m0} = \frac{2(8)}{4.5} = 3.556\ \text{mS}, \qquad g_m = 3.556\left(1 - \frac{-1.441}{-4.5}\right) = 2.42\ \text{mS}$$

Stage 2 (unloaded): $A_{v2} = -g_m R_D = -(2.417\ \text{mS})(2.2\ \text{k}\Omega) = \mathbf{-5.32}$.

Stage 1 (loaded by $R_{G2} = 10$ MΩ): $A_{v1} = -g_m(2.2\text{k} \parallel 10\text{M}) = \mathbf{-5.32}$ (the load changes it only in the 4th figure).

Overall: $A_v = A_{v1}A_{v2} = (-5.32)(-5.32) = \mathbf{28.3}$, in phase.

Output: $V_o = A_v V_i = 28.3 \times 20\ \text{mV} = \mathbf{565\ mV}$ (0.57 V).

Also $Z_i = 10$ MΩ, $Z_o = 2.2$ kΩ.

New questions

N1.

$I_{DQ} = k(V_{GSQ} - V_{GS(\text{Th})})^2 = 0.3\ \text{mA/V}^2 \times (3\ \text{V})^2 = \mathbf{2.7\ mA}$.

$g_m = 2k(V_{GSQ} - V_{GS(\text{Th})}) = 2(0.3)(3) = \mathbf{1.8\ mS}$.

Check: $V_{ov} = 3$ V, $g_m = 2I_D/V_{ov} = 2(2.7)/3 = 1.8$ mS. Same.

N2.

a. Self-bias: $V_{GS} = -1500 I_D$. Solve $I_D = 12\ \text{mA}(1 - 1500 I_D/4)^2$. Roots 1.67 mA and 4.25 mA; the second gives $V_{GS} = -6.38\ \text{V} < V_P$, reject.

$I_{DQ} = \mathbf{1.67\ mA}$, $V_{GSQ} = \mathbf{-2.51\ V}$.

$g_{m0} = 2(12)/4 = 6$ mS, $g_m = 6(1 - 2.507/4) = \mathbf{2.24\ mS}$.

b. $Z_i = R_G = \mathbf{1\ M\Omega}$. $Z_o = R_S \parallel 1/g_m = 1.5\text{k} \parallel 446.6 = \mathbf{344\ \Omega}$.

$$A_v = \frac{g_m R_S}{1 + g_m R_S} = \frac{3.359}{4.359} = \mathbf{0.771}.$$

c. $R_S \parallel R_L = 600\ \Omega$, $g_m(600) = 1.344$:

$$A_{v_L} = \frac{1.344}{2.344} = \mathbf{0.573}.$$

$A_{v_s} = \dfrac{1\text{ M}}{1\text{ M} + 10\text{ k}} \times 0.573 = \mathbf{0.568}$. R_{sig} barely matters; R_L matters a lot because it is comparable with Z_o.

N3.

a. $Z_i = R_S \parallel 1/g_m = 1.2\text{k} \parallel 250 = \mathbf{207\ \Omega}$. $Z_o = R_D = \mathbf{3.3\ k\Omega}$.

b. $A_{v_{NL}} = g_m R_D = 4 \times 3.3 = 13.2$.

$A_{v_L} = g_m(R_D \parallel R_L) = 4\text{ mS} \times 2.481\text{ k}\Omega = \mathbf{9.92}$.

$A_{v_s} = \dfrac{207}{207 + 200} \times 9.92 = \mathbf{5.05}$.

c. The CG input resistance (207 Ω) is about equal to R_{sig} (200 Ω), so the input divider passes only about half of V_s.

N4.

a. $V_{GS} = V_P(1 - \sqrt{I_D/I_{DSS}}) = -5(1 - \sqrt{0.4}) = \mathbf{-1.84\ V}$.
$R_S = 1.838/4\text{ mA} = \mathbf{459\ \Omega}$ (use 470 Ω standard).

b. $g_{m0} = 2(10)/5 = 4$ mS, $g_m = 4(1 - 1.838/5) = \mathbf{2.53\ mS}$.

c. $r_d = 1/25\ \mu\text{S} = 40$ kΩ. $R_D \parallel r_d = 10/2.53\text{ mS} = 3.953$ kΩ.

$R_D = \dfrac{1}{1/3.953\text{k} - 1/40\text{k}} = \mathbf{4.39\ k\Omega}$.

With 4.3 kΩ: $A_v = -2.53\text{ mS} \times (4.3\text{k} \parallel 40\text{k}) = -9.82$. With 4.7 kΩ: $A_v = -10.6$. Pick 4.3 kΩ if the gain must not exceed 10, or 4.7 kΩ if it must reach at least 10.

d. $V_{DS} = 30 - 4\text{ mA}(4.386 + 0.459)\text{ k}\Omega = 10.6$ V. Needs $V_{DS} > V_{GS} - V_P = 3.16$ V. OK.

(Note: with $V_{DD} = 20$ V, V_{DS} would be only 0.62 V and the design would fail. Always recheck.)

e. $A_v = \dfrac{-g_m R_D}{1 + g_m R_S + (R_D + R_S)/r_d} = \dfrac{-(2.53)(4.386)}{1 + 1.162 + 0.121} = \mathbf{-4.86}$.

N5.

a. Drain feedback: $V_{GS} = V_{DS} = 15 - 2200 I_D$. Solve $I_D = 0.5\text{ mA}(13 - 2200 I_D)^2$: roots 4.54 mA and 7.69 mA. The second gives $V_{GS} = -1.92$ V, below threshold, reject.

$I_{DQ} = \mathbf{4.54\ mA}$, $V_{GSQ} = \mathbf{5.01\ V}$, $g_m = 2(0.5)(5.013 - 2) = \mathbf{3.01\ mS}$.

b. Stage 1 load: $R_F \parallel R_D \parallel Z_{i2} = 12\text{M} \parallel 2.2\text{k} \parallel 1\text{M} \approx 2.195$ kΩ.

$A_{v1} = -g_m(2.195\ \text{k}\Omega) = \mathbf{-6.61}$.

$Z_{i1} = \dfrac{R_F}{1 + g_m(R_D \parallel R_{L1})} = \dfrac{12\ \text{M}}{1 + 6.61} = \mathbf{1.58\ M\Omega}$ (1.57 MΩ using R_D alone).

c. $A_{v2} = \dfrac{g_m(R_S \parallel R_L)}{1 + g_m(R_S \parallel R_L)} = \mathbf{0.573}$ (from N2c).

$A_v = A_{v1}A_{v2} = (-6.61)(0.573) = \mathbf{-3.79}$.

$V_i = \dfrac{1.576}{1.576 + 0.5} \times 50\ \text{mV} = 37.96$ mV, so $V_o = -3.79 \times 37.96 = \mathbf{-144\ mV}$ (144 mV, 180° out of phase). Overall $A_{v_s} = -2.88$.

R_{sig} matters here because the drain-feedback Z_i (1.6 MΩ) is not much bigger than 500 kΩ.

d. $Z_o = Z_{o2} = R_S \parallel 1/g_m = \mathbf{344\ \Omega}$.

N6.

a. $V_{ov} = \sqrt{\dfrac{2I_D}{\mu_n C_{ox}(W/L)}} = \sqrt{\dfrac{1}{8}} = \mathbf{0.354\ V}$. $V_{GS} = 0.6 + 0.354 = \mathbf{0.954\ V}$.

$g_m = 2I_D/V_{ov} = 1/0.354 = \mathbf{2.83\ mA/V}$.

b. $k = \frac{1}{2}(0.2)(40) = \mathbf{4\ mA/V^2}$. $2kV_{ov} = 2(4)(0.354) = 2.83$ mA/V. Same.

c. $A_v = -g_m(R_D \parallel R_L) = -2.83 \times 3.33 = \mathbf{-9.43}$.

6. Quick recall check

Question

1. $Z_o = R_S \parallel 1/g_m$.
2. $Z_i \approx R_F/(1 + g_m R_D)$.
3. $Z_i \approx 1/g_m = 200\ \Omega$.
4. $g_m = 2I_D/V_{ov} = 2(2)/0.4 = 10$ mA/V.
5. $A_v = (-4)(-4) = +16$, in phase.